

THE NORMAL CLOSURE OF CLASSICAL BRAID GROUPS IN TORUS BRAID GROUPS

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1. PRELIMINARIES

1.1. Definitions.

Definition 1.1.1 (Configuration Spaces). Let S be a surface. We define the *ordered configuration space of S* as

$$\widehat{C}_n(S) = \left\{ x = (x_1, x_2, \dots, x_n) \in \prod_{i=1}^n \text{int}(S) : x_i = x_j \Rightarrow i = j \right\},$$

endowed with the subspace topology of the product topology on $\prod_{i=1}^n S$. Note that the symmetric group Σ_n acts on $\widehat{C}_n(S)$ by permuting the entries of each x . We define the *configuration space of S* , $C_n(S)$ as the orbit space $\widehat{C}_n(S)/\Sigma_n$. It is worth mentioning that the projection $\widehat{C}_n(S) \rightarrow C_n(S)$ is a regular covering with Deck transformation group Σ_n .

This definition simply abstracts the idea of choosing n distinct points on S , where the former keeps track of the order, while latter cares only about which points we choose. When $S = D^2$, we simply write $\widehat{C}_n$ and C_n for its configuration spaces.

Definition 1.1.2 (Surface Braid Groups). Let $x \in C_n(S)$. We define the *geometric braid group on S based at x* as

$$B_n(S) := \pi_1(C_n(S), x),$$

and the *geometric pure braid group*

$$P_n(S) := \pi_1(\widehat{C}_n(S), x).$$

Alternatively, note that we have a map $\varphi : B_n(S) \rightarrow \Sigma_n$. We can also define $P_n(S) = \ker \varphi$.

1.2. Classical Braid Groups. There are three equivalent definitions of B_n , the classical braid group.

- (1) $B_n := B_n(D^2)$.
- (2) $B_n := \text{MCG}(D^2, \{x_1, \dots, x_n\})$.
- (3) $B_n := \langle \sigma_1, \dots, \sigma_{n-1} \mid \sigma_{i+1}\sigma_i\sigma_{i+1} = \sigma_i\sigma_{i+1}\sigma_i, \sigma_i\sigma_j = \sigma_j\sigma_i \mid i - j \geq 2 \rangle$.

We similarly define $PB_n := \ker(B_n \rightarrow \Sigma_n)$.

1.3. Embeddings. Let $f : X \rightarrow Y$ be an injection of surfaces. We get an induced map

$$\begin{aligned} \widehat{f} : C_n(X) &\rightarrow C_n(Y) \\ [x_1, \dots, x_n] &\mapsto [f(x_1), \dots, f(x_n)], \end{aligned}$$

which in turn induces a homomorphism $\widehat{f}_* : B_n(X) \rightarrow B_n(Y)$.

Theorem 1.3.1 (Paris, Rolfsen). Let $f : X \hookrightarrow Y$ be inclusion. Suppose that $X \neq S^2, \mathbb{RP}^2$, and that none of the components of $\overline{Y \setminus X}$ is D^2 . Then $\widehat{f}_* : B_n(X) \rightarrow B_n(Y)$ is injective.

Corollary 1.3.2. Let $x_1, \dots, x_n$ be distinct points in T^2 , and let $\{x_1, \dots, x_n\} \subset \text{int}(D) \subset T^2$, where D is an embedded disc. Then $B_n(D) \leq B_n(T^2)$.

2. TORUS BRAIDS

Given S , consider the inclusion $i : \widehat{C}_n(S) \hookrightarrow \prod^n S$, which induces $i_* : P_n(S) \rightarrow \prod^n \pi_1(S)$. Birman showed that $\ker i_* = [P_n(T^2), P_n(T^2)]$ (hence $P_n(T^2)^{\text{ab}} \cong \prod^n \pi_1(T^2)$). It was later shown by Goldberg that $P_n(D)^{P_n(T^2)} = \ker i_*$. Therefore

$$P_n(D)^{P_n(T^2)} = [P_n(T^2), P_n(T^2)].$$

2.1. Generalizing to $B_n(T^2)$. Using that $T^2 \cong S^1 \times S^1$ is a topological abelian group, write $e \in T^2$ for the identity.

Definition 2.1.1.

$$\widehat{C}_n^e(T^2) := \left\{ (x_1, \dots, x_n) \in \widehat{C}_n(T^2) : \prod x_i = e \right\}.$$

$$C_n^e(T^2) := \left\{ [x_1, \dots, x_n] \in C_n(T^2) : \prod x_i = e \right\}.$$

Note that because T^2 is abelian, $C_n^e(T^2)$ is well-defined. We then define

$$\begin{aligned} P_n^e(T^2) &:= \pi_1 \widehat{C}_n^e(T^2) \\ B_n^e(T^2) &:= \pi_1 C_n^e(T^2). \end{aligned}$$

Note that we have $P_n^e(T^2) \hookrightarrow B_n^e(T^2)$.

Proposition 2.1.2.

$$\begin{aligned} P_n(D) &\leq P_n^e(T^2) \\ B_n(D) &\leq B_n^e(T^2). \end{aligned}$$

Proof. Suffices to show that the generators $\sigma_i \in B_n(D)$ are in $\pi_1 C_n^e(T^2)$. ■

Theorem 2.1.3.

$$B_n(D)^{B_n(T^2)} = B_n^e(T^2).$$

Lemma 2.1.4. *The map*

$$\begin{aligned} \rho : C_n(T^2) &\rightarrow T^2 \\ [x_1, \dots, x_n] &\mapsto \prod x_i \end{aligned}$$

is a fiber bundle with fiber $C_n^e(T^2)$.

Corollary 2.1.5. *We have a short exact sequence*

$$1 \rightarrow B_n^e(T^2) \rightarrow B_n(T^2) \xrightarrow{\rho_*} \mathbb{Z}^2 \rightarrow 1.$$

Analogously, we obtain the short exact sequence

$$1 \rightarrow P_n^e(T^2) \rightarrow P_n(T^2) \rightarrow \mathbb{Z}^2 \rightarrow 1.$$

Using a known presentation for $B_n(T^2)$ in terms of half-twist generators $\sigma_1, \dots, \sigma_{n-1}$ for $B_n(D)$ and generators a, b for $\pi_1(T^2)$, the map ρ_* is given by

$$\begin{aligned} a &\mapsto (1, 0) \\ b &\mapsto (0, 1) \\ \sigma_i &\mapsto (0, 0). \end{aligned}$$

Lemma 2.1.6.

$$B_n(T^2)^{\text{ab}} \cong \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}_2$$

Proof. Abelianize the presentation. ■

Lemma 2.1.7.

$$B_n^e(T^2) = [B_n(T^2), B_n(T^2)] \cdot B_n(D).$$

Proof. The following diagram commutes.

$$\begin{array}{ccccccc}
& & & 1 & & & \\
& & & \downarrow & & & \\
& & & B_n^e(T^2) & & & \\
& & & \downarrow & & & \\
1 & \longrightarrow & [B_n(T^2), B_n(T^2)] & \longrightarrow & B_n(T^2) & \longrightarrow & \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}_2 \longrightarrow 1 \\
& & & & \downarrow \rho_* & \swarrow & \\
& & & & B_n(T^2) & & \\
& & & & \downarrow & & \\
& & & & 1 & &
\end{array}$$

Diagram chasing proves $[B_n(T^2), B_n(T^2)] \leq B_n^e(T^2)$. Since $B_n(D) \leq B_n^e(T^2)$, we get $[B_n(T^2), B_n(T^2)] \cdot B_n(D) \leq B_n^e(T^2)$. On the other hand, given $[B_n(T^2), B_n(T^2)] \leq B_n^e(T^2) \leq B_n(T^2)$, third isomorphism theorem tells us that

$$\begin{aligned}
\frac{B_n(T^2)/[B_n(T^2), B_n(T^2)]}{B_n^e(T^2)/[B_n(T^2), B_n(T^2)]} &\cong \frac{\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}/2}{B_n^e(T^2)/[B_n(T^2), B_n(T^2)]} \\
&\cong B_n(T^2)/B_n^e(T^2) \\
&\cong \mathbb{Z} \times \mathbb{Z}
\end{aligned}$$

whence $B_n^e(T^2)/[B_n(T^2), B_n(T^2)] \cong \mathbb{Z}/2$.

Since $\sigma_1 \in B_n^e(T^2)$ and $\sigma_1 \notin [B_n(T^2), B_n(T^2)]$, σ_1 represents the non-zero coset. Then

$$B_n^e(T^2) = [B_n(T^2), B_n(T^2)] \cup [B_n(T^2), B_n(T^2)] \cdot \sigma_1 \leq [B_n(T^2), B_n(T^2)] \cdot B_n(D).$$

■

We are now ready to prove the main theorem.

Proof. The following facts will be necessary.

- (1) $B_n(D) \leq B_n^e(T^2)$.
- (2) $B_n^e(T^2) = [B_n(T^2), B_n(T^2)] \cdot B_n(D)$.
- (3) $B_n(T^2) = P_n(T^2) \cdot B_n(D)$.
- (4) $[P_n(T^2), P_n(T^2)] = P_n(D)^{P_n(T^2)} \leq B_n(D)^{B_n(T^2)}$.

By Fact 1, and since $B_n^e(T^2) \leq B_n(T^2)$, then $B_n(D)^{B_n(T^2)} \leq B_n^e(T^2)$.

For the reverse inclusion, we use Fact 2. Clearly $B_n(D) \leq B_n(D)^{B_n(T^2)}$, so it suffices to show $[B_n(T^2), B_n(T^2)] \leq B_n(D)^{B_n(T^2)}$.

Let $\alpha\beta\alpha^{-1}\beta^{-1} \in [B_n(T^2), B_n(T^2)]$, and consider the quotient map

$$B_n^e(T^2) \xrightarrow{q} B_n^e(T^2)/B_n(D)^{B_n(T^2)}.$$

By Fact 3, we can write $\alpha = s_1 t_1$ and $\beta = s_2 t_2$, where $s_i \in P_n(T^2)$ and $t_i \in B_n(D)$. Hence $\alpha\beta\alpha^{-1}\beta^{-1} = s_1 t_1 s_2 t_2 t_1^{-1} s_1^{-1} t_2^{-1} s_2^{-1}$. Since $B_n(D) \leq B_n(D)^{B_n(T^2)}$, then $t_1, t_2 \in \ker(q)$. Thus $q(\alpha\beta\alpha^{-1}\beta^{-1}) = q(s_1 s_2 s_1^{-1} s_2^{-1})$.

We now use Fact 4. We have $s_1 s_2 s_1^{-1} s_2^{-1} \in [P_n(T^2), P_n(T^2)] = P_n(D)^{P_n(T^2)} \leq B_n(D)^{B_n(T^2)}$, hence $\alpha\beta\alpha^{-1}\beta^{-1} \in \ker(q) = B_n(D)^{B_n(T^2)}$. Therefore $[B_n(T^2), B_n(T^2)] \leq B_n(D)^{B_n(T^2)}$. ■

REFERENCES

- [1] Liming Pang. *A Geometric Interpretation of the Normal Closure of the Braid Group B_n in the braid group of the torus $B_n(T)$* . 2019. arXiv: 1806.08000 [math.GT]. URL: <https://arxiv.org/abs/1806.08000>.